

PHASE 3

Project Design Phase

AI StudyBuddy – AI-Powered Learning and Study Support Platform

Project Focus
AI-assisted learning resources, secure backend services, and personalized study support.

Phase Objective

To define the structure of the application, its major components, data relationships, software architecture, and interaction flow before implementation.

3.1 Overall Architecture

AI StudyBuddy uses a modular REST-based backend. Requests from the client reach the Express server, pass through security and routing layers, interact with business logic and database models, and invoke the Gemini service when AI processing is required.

3.2 Major Components

- Client application: a React-based frontend through which users authenticate, submit material, and request learning resources.
- Express server: receives HTTP requests, applies middleware, selects routes, and returns responses.
- Authentication middleware: checks JWT credentials and enforces protected access.
- Route handlers: organize endpoints for authentication, materials, AI functions, and administrative operations.
- AI service layer: communicates with Google Gemini and formats AI-generated learning content.
- Database layer: uses MongoDB and Mongoose to persist user data, study material, and generated resources.

3.3 Data Model

- User: stores identity, email, hashed password, role, and creation information.
- Study Material: stores the owner, title, subject, content, and creation time.
- Summary: connects a user and source material with an AI-generated summary.
- Flashcard: links the user and source material with a question and answer.
- Quiz: connects the source material and user with generated questions.
- Study Plan: stores the user's generated plan together with the relevant examination date.

3.4 Key Data Relationships

- One user can own multiple study materials.
- A study material can have an associated summary.
- A single study material can produce several flashcards.
- A study material can produce a quiz containing multiple questions.
- A user can create several study plans for different subjects or schedules.
- All generated resources remain associated with the user who created them.

3.5 MVC Organization

The backend follows the MVC pattern. Models handle MongoDB structures and validation, controllers contain application logic, and the API response layer returns structured JSON for the React client. This separation supports maintainability, debugging, and future feature additions.

3.6 User Flow

A typical interaction begins with registration or login. After authentication, the student reaches the dashboard, supplies study content, chooses an AI function, receives the generated resource, and can retain it for later revision.